

Spring 2026 - Claudeversion#1

Noetic Learning Mathematics Contest

Grade 3

Practice Paper

45 minutes • No calculators allowed

Student Name: _____

For Teacher Use Only

Number of correct answers: _____

Score (Number of correct answers $\times$ 5): _____

1) $35 + 47 = ?$

Answer: _____

2) Emma has 5 shelves of books. Each shelf holds 8 books. She donates 12 books to the library. How many books does Emma have left?

Answer: _____ books

3) A farmer plants tomato seeds in rows. He plants 6 seeds in the first row, 10 seeds in the second row, 14 seeds in the third row, and so on. How many seeds does he plant in the fifth row?

Answer: _____ seeds

4) Maya has 4 bags of apples. Each bag contains 9 apples. She eats 2 apples every day. How many days will it take Maya to eat all the apples?

Answer: _____ days

5) A store sells pencils for \$2 each and notebooks for \$5 each. Leo has \$23. He buys 2 notebooks. How many pencils can he buy with the money he has left?

Answer: _____ pencils

6) A rectangular garden has a length of 12 feet and a width of 8 feet. What is the perimeter of the garden?

Answer: _____ feet

7) Today is Monday. Sofia's soccer tournament starts in 18 days. What day of the week will the tournament start?

Answer: _____

8) A school bus picks up students at 3 stops. At the first stop, 7 students get on. At the second stop, 5 students get on and 3 get off. At the third stop, 4 students get on and 2 get off. How many students are on the bus after the third stop?

Answer: _____ students

9) There are 24 students in a class. One-third of the students have a dog, and one-fourth of the students have a cat. How many students have either a dog or a cat? (Assume no student has both.)

Answer: _____ students

10) Ben has some trading cards. He gives half of them to his brother. Then he gives 8 cards to his friend. He now has 10 cards left. How many cards did Ben start with?

Answer: _____ cards

11) A clock reads 3:40 PM right now. A movie starts in 1 hour and 35 minutes. What time does the movie start?

Answer: _____ PM

12) Four friends — Ana, Bo, Cal, and Dan — each hold two cards showing the same number. Some cards face backward. Ana holds 5, Bo holds 8, and Dan holds 3. The sum of all eight cards is 36. What number does Cal hold on each card?

Answer: _____

13) A bakery makes 15 muffins every hour. It opens at 7:00 AM and by 10:00 AM all muffins have been sold. If muffins are sold in boxes of 5, how many boxes were sold in total?

Answer: _____ boxes

14) Zoe, Mia, and Lily together have 45 stickers. Zoe has 3 more stickers than Mia, and Mia has 3 more stickers than Lily. How many stickers does Zoe have?

Answer: _____ stickers

15) In a jar there are red, blue, and green marbles. There are 5 red marbles. The number of blue marbles is twice the number of red marbles. The number of green marbles is 3 more than the number of blue marbles. How many marbles are in the jar in total?

Answer: _____ marbles

16) A snail climbs 4 feet up a wall during the day but slides back 1 foot each night. The wall is 12 feet tall. How many days will it take the snail to reach the top?

Answer: _____ days

17) Using the digits 1, 2, 3, 4, and 6 (each used exactly once), fill in the boxes to make an addition problem with the largest possible sum: $\blacksquare\blacksquare\blacksquare + \blacksquare\blacksquare = ?$ What is the largest possible sum?

Answer: _____

18) There are 20 children at a party. Each child shakes hands with every other child exactly once. How many handshakes happen in total?

Answer: _____ handshakes

19) A number is called 'lucky' if it is divisible by both 2 and 3. How many lucky numbers are there from 1 to 36?

Answer: _____ lucky numbers

20) Eight teams enter a single-elimination tournament (one loss and you're out). Every game must have a winner. How many games are played in total before one champion is crowned?

Answer: _____ games

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Answers & Solutions

1) Answer: 82

Solution: Add the tens: $30 + 40 = 70$. Add the ones: $5 + 7 = 12$. Combine: $70 + 12 = 82$.

2) Answer: 28 books

Solution: Emma starts with $5 \times 8 = 40$ books. She donates 12, so she has $40 - 12 = 28$ books left.

3) Answer: 22 seeds

Solution: The pattern increases by 4 each row: 6, 10, 14, 18, 22. The 5th row has 22 seeds.

4) Answer: 18 days

Solution: Total apples: $4 \times 9 = 36$. At 2 per day: $36 \div 2 = 18$ days.

5) Answer: 6 pencils

Solution: Two notebooks cost $2 \times \$5 = \10 . Money left: $\$23 - \$10 = \$13$. Number of \$2 pencils: $\$13 \div \$2 = 6$ remainder \$1, so Leo can buy 6 pencils.

6) Answer: 40 feet

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (12 + 8) = 2 \times 20 = 40$ feet.

7) Answer: Friday

Solution: Starting from Monday, count 18 days forward. $18 = 2 \text{ weeks} + 4 \text{ days}$. Monday + 4 days = Friday.

8) Answer: 11 students

Solution: After stop 1: 7. After stop 2: $7 + 5 - 3 = 9$. After stop 3: $9 + 4 - 2 = 11$ students.

9) Answer: 14 students

Solution: Dogs: $24 \div 3 = 8$ students. Cats: $24 \div 4 = 6$ students. Total (no overlap): $8 + 6 = 14$ students.

10) Answer: 36 cards

Solution: Work backwards. Before giving to friend: $10 + 8 = 18$ cards. 18 was half of the original, so Ben started with $18 \times 2 = 36$ cards.

11) Answer: 5:15 PM

Solution: $3:40 \text{ PM} + 1 \text{ hour} = 4:40 \text{ PM}$. $4:40 \text{ PM} + 35 \text{ minutes} = 5:15 \text{ PM}$.

12) Answer: 2

Solution: Known cards total: Ana ($5+5=10$) + Bo ($8+8=16$) + Dan ($3+3=6$) = 32. Cal's two cards must sum to $36 - 32 = 4$. Since both of Cal's cards show the same number, each card shows $4 \div 2 = 2$.

13) Answer: 9 boxes

Solution: From 7:00 AM to 10:00 AM = 3 hours. Muffins made: $3 \times 15 = 45$. Boxes sold: $45 \div 5 = 9$ boxes.

14) Answer: 18 stickers

Solution: Let Lily = x . Then Mia = $x+3$, Zoe = $x+6$. Total: $x + (x+3) + (x+6) = 45 \rightarrow 3x + 9 = 45 \rightarrow 3x = 36 \rightarrow x = 12$. Zoe has $12 + 6 = 18$ stickers.

15) Answer: 28 marbles

Solution: Red = 5. Blue = $2 \times 5 = 10$. Green = $10 + 3 = 13$. Total = $5 + 10 + 13 = 28$ marbles.

16) Answer: 4 days

Solution: Each full day-night cycle the snail gains a net 3 feet (climbs 4, slides 1). After day 1 night: 3 ft. After day 2 night: 6 ft. After day 3 night: 9 ft. On day 4, the snail starts at 9 ft and climbs 4 ft, reaching 13 ft — it passes 12 ft during day 4 and stops there. Answer: 4 days.

17) Answer: 673

Solution: To maximize a 3-digit + 2-digit sum, put the three largest digits (6, 4, 3) in the 3-digit number and the two smallest (1, 2) in the 2-digit number. Best arrangement: $641 + 32 = 673$ (or $642 + 31 = 673$). The largest possible sum is 673.

18) Answer: 190 handshakes

Solution: Each of 20 children shakes hands with 19 others: $20 \times 19 = 380$. Each handshake is counted twice, so total = $380 \div 2 = 190$ handshakes.

19) Answer: 6 lucky numbers

Solution: A number divisible by both 2 and 3 must be divisible by 6. Multiples of 6 up to 36: 6, 12, 18, 24, 30, 36. That is 6 lucky numbers.

20) Answer: 7 games

Solution: In a single-elimination tournament every game eliminates exactly 1 team. To get from 8 teams to 1 champion, 7 teams must be eliminated. Therefore, 7 games are played.

• Quick Answer Reference •

Q#	Answer	Q#	Answer
1	82	2	28 books
3	22 seeds	4	18 days
5	6 pencils	6	40 feet
7	Friday	8	11 students
9	14 students	10	36 cards
11	5:15 PM	12	2
13	9 boxes	14	18 stickers

15	28 marbles	16	4 days
17	673	18	190 handshakes
19	6 lucky numbers	20	7 games